

CORP1 — Product Spec & Build Plan

2026-09-17 · @Someone

Executive Summary

CORP1 (Creator Opportunity Research Platform) is a research and opportunity-discovery system for creating digital products with existing creators.

CORP1 is not an influencer database. It is an **opportunity pipeline**:

Creator → Audience Problem → Evidence → Product Opportunity → Human Decision

The Core Question

Which creators already have an audience with a recurring problem we can turn into a product, and what evidence shows that product is worth building?

Scope

CORP1 covers research only — steps 1 through 9 of the full CORP vision.

In Scope (CORP1)	Out of Scope (CORP2)
Find creators (10K–200K followers)	Contact creators
Research content and audience automatically	Partnership proposals
Detect recurring audience problems	Deal tracking
Generate product ideas from problems	Product building
Validate demand with external evidence	Sales and revenue
Rank opportunities by evidence strength	Marketplace integration
Present dossier for human decision	Partnership management

Design Principle

CORP1 eliminates manual research. The machine collects and analyses evidence across multiple platforms. You remain the decision-maker at the important gates.

Stage 3: Product Specification (Frozen)

This section freezes CORP1's complete-system scope before Stage 4 (Architecture Gate), following the same AI-assisted development workflow used for the Acquisition system. Vertical slices in the build phases below are an implementation technique only — they do not reduce this frozen scope.

Complete Product Flow (Frozen)

Trend Scan → Niche Discovery (recursive drill-down) → Creator Discovery → Creator Profile → Content and Audience Research → Problem Clustering → Product Idea Generation → Demand Validation and Ranking → Dossier → Human Decision Gate (Reject / Research More / Watch / Approve) → on Approve: handoff to CORP2.

Core Product Capabilities (Frozen)

- Autonomous topic discovery from Google Trends — no user input required to start a discovery pass
- User-suggested niches accepted as an alternate entry point alongside autonomous discovery
- Recursive niche drill-down until a leaf niche is specific enough to name a concrete digital product
- Multi-source evidence capability providers: ProblemProvider, SearchIntentProvider, TrendProvider, PlanningIntentProvider, TransactionProvider, SolutionProvider, MonetisationProvider, DissatisfactionProvider
- Creator discovery within a qualified niche (10K–200K followers, multi-platform)
- Evidence-based opportunity scoring
- Decision-ready dossier generation
- Research registry with dedup and re-scan scheduling
- Human decision gate with four states: Reject / Research More / Watch / Approve
- Full evidence provenance (source → evidence type → observation vs. inference) through the entire pipeline

Niche Exclusion Rules (Frozen)

Excluded regardless of demand signal, dropped at niche-discovery time before creator discovery begins — never surfaced for review:

- Illegal in target markets — drugs, weapons, counterfeit goods, piracy, unlicensed regulated activity
- Adult content, gambling, tobacco, alcohol-focused niches

- Political and religious advocacy or activism content
- Regulated advice niches — medical, legal, financial — where a wrong digital product could cause real harm

Research Registry (Frozen)

- Every topic, niche, and sub-niche researched is logged with: date researched, sources queried, drill depth reached, evidence count
- A niche becomes eligible for re-scan after **90 days**
- Re-scanning re-queries all evidence sources and updates the existing registry entry — it never creates a duplicate niche record
- The registry is checked before every discovery pass so CORP1 does not repeat research on a niche still inside its 90-day window

Human Decision Gate (Frozen — Four States)

Every dossier ends at exactly one of four states:

- **Reject** — not a strong enough opportunity. The niche/creator stays in the registry as researched and is not re-suggested outside normal re-scan timing.
- **Research More** — CORP1 automatically drills one level deeper into the niche and re-queries all evidence sources for that specific niche, then produces an updated dossier.
- **Watch** — parked. Automatically re-scored on the same 90-day re-scan cycle as the registry, so it can resurface on its own if the evidence strengthens.
- **Approve** — moves to CORP2; triggers the handoff package below.

CORP1 → CORP2 Handoff Package (Frozen)

On Approve, CORP1 produces a decision package containing:

- The full dossier
- The complete evidence trail — every evidence record and source link the dossier was built from
- The full niche drill-down path (e.g. Automotive → Track Builds → Suspension) showing how specific the opportunity is
- Your review notes from the decision gate

This mirrors the Acquisition system's dossier-snapshot pattern: CORP2 references this package rather than querying the CORP1 research database directly.

Out of Scope (Frozen)

CORP1 never contacts creators, drafts or sends outreach messages, negotiates terms or revenue splits, or builds or sells the product. All of this is CORP2 territory, triggered only after Approve.

Provenance Invariant (Frozen)

CORP1 EVIDENCE $\neq$ HUMAN OBSERVATION $\neq$ AI INFERENCE. Every claim in a dossier — a problem, a score, a recommendation — must trace back to which evidence provider collected it, which raw source it came from, and whether it is a direct observation from the data or an LLM inference drawn from that observation. The system must never silently present its own inference as raw evidence.

Stage 4: Architecture Gate (Frozen)

Stage 4 is completed by defining the system boundary, the authoritative data model, state machines, the capability interface freeze, the automation/human-gate matrix, and acceptance tests — building on CORP1's existing SQLAlchemy models rather than replacing them. Coding agents implement this frozen architecture; they do not redesign it while coding.

Frozen System Boundary

Governing rule: **CORP1 discovers and evaluates opportunities; CORP2 pursues and manages the creator relationship.**

CORP1 owns topic scanning, the niche tree, evidence, scoring, dossiers, and the four-state decision gate. CORP2 owns everything after Approve — outreach, negotiation, partnership, product build. Both run as separate applications sharing one Postgres instance. CORP2's tables are namespaced (Acq*) and reference CORP1 only through the Dossier handoff — CORP2 never queries CORP1's niche, evidence, or candidate tables directly.

Frozen Data Model — Modifications to Existing Models

Niche (existing table, modified):

- Add `parent_niche_id` (self-referential FK, nullable) — becomes the source of truth for tree structure; the existing `parent_domain` string stays as a display label only
- Add `depth` (integer, 0 = broad topic) — enforces the max-3-levels drill cap
- Existing `last_researched_at` / `next_recheck_at` become the research-registry fields directly: `next_recheck_at = last_researched_at + 90 days` on every scan

- Existing `lifecycle_status` enum gains one value: `excluded` — set immediately when a niche matches a Stage 3 exclusion rule, so it is dropped before creator discovery and never resurfaces in review

NicheCandidate (existing table, modified):

- Add `parent_candidate_id` (self-referential FK, nullable) — the specific drill-down edge this candidate was drilled from
- Add `depth` (integer) — mirrors Niche.depth
- Existing `naming_method` / `naming_terms` / `naming_confidence` / `evidence_count` / `source_count` already satisfy "niches come from evidence, not imagination" — unchanged
- Existing `status` enum gains one value: `drilling` — set while a candidate is mid-recursion, so a partial pass is distinguishable from a finished one

Evidence (existing table, modified):

- Add `evidence_type` enum: `problem` | `search_intent` | `trend` | `planning_intent` | `transaction` | `solution` | `monetisation` | `dissatisfaction` — maps 1:1 to the eight capability-provider interfaces in the Capability Interface Architecture section
- Add `origin` enum: `observation` | `inference` — `observation` for raw data pulled directly from a source, `inference` for anything an LLM derived from it (a cluster label, a niche name, a problem summary). This makes the Provenance Invariant concrete: every Evidence row is one or the other, never ambiguous
- Existing `source_type` / `source_platform` / `raw_text` / `access_method` / `compliance_status` are unchanged

HumanDecision (existing table, modified):

- `decision` enum (currently `approve` | `reject` | `watch`) gains a 4th value: `research_more` — completes the Stage 3 four-state gate
- Add `dossier_id` (FK) — the gate now decides on a Dossier, not a bare `creator_id`
- On `research_more`: a new `ResearchRun` is created with `run_type` = `niche_discovery`, scoped to the specific `NicheCandidate` at `depth + 1`
- On `watch`: no new column needed — a scheduled job re-scores any Dossier whose Niche `next_recheck_at` has passed, reusing the same 90-day registry cycle

New model — Dossier (table `dossiers`, does not exist today):

- `id`, `creator_id` (FK), `niche_id` (FK), `opportunity_score_id` (FK), `research_run_id` (FK)
- `content` (JSONB) — the rendered dossier sections shown in the Example Dossier Summary above

- `dossier_evidence` join table — the complete evidence trail for the CORP2 handoff package
- `niche_path` — the full drill-down path (derived by walking `Niche.parent_niche_id`), included in the handoff package
- `generated_at`, `superseded_at` (nullable) — same re-scan versioning pattern as `CreatorScore/OpportunityScore`
- `status` enum: `pending_review` | `approved` | `rejected` | `watching` | `research_more_in_progress` — a denormalized mirror of the latest `HumanDecision` for fast dashboard queries; `HumanDecision` remains the append-only source of truth

Frozen State Machines

Machine	States
Niche drill lifecycle	candidate → excluded (if matched) / active → expand (drilling to depth+1) → recheck_due (past next_recheck_at) → active
NicheCandidate lifecycle	staged → drilling (not yet specific enough) → staged (at new depth) → promoted (specific enough) / merged / rejected
Dossier lifecycle	pending_review → approved (CORP2 handoff) / rejected / watching (auto-rescore loop) / research_more_in_progress (new ResearchRun → new Dossier version, old superseded_at set)

Automation / Human-Gate Matrix (Frozen)

Action	Authority
Topic scan (Google Trends pull)	AUTO
Evidence collection (all 8 provider types)	AUTO
Niche naming / synthesis from evidence	AI-AUTO with provenance (origin = inference)
Drill-down depth decision ("specific enough?")	AI-AUTO with provenance
Niche exclusion matching	AUTO — deterministic rule match against the Stage 3 exclusion list, never an LLM judgment call

Action	Authority
Creator discovery within a qualified niche	AUTO
Opportunity / creator scoring	AUTO (rules) + AI-AUTO (LLM-assisted components), confidence-banded
Dossier generation	AI-AUTO with provenance
Reject / Research More / Watch / Approve	HUMAN
Handoff package creation	AUTO, triggered by Approve
Un-excluding a niche	HUMAN — never automatic

Automation Failure Rule (Frozen)

Raw evidence must survive automation failure. Safe retries are permitted. If a stage fails repeatedly, the ResearchRun records `status = failed`, visible on the campaign dashboard. CORP1 has no push-notification/attention engine — it is a pull-based research dashboard you check, not a system that pages you, so a failed run surfaces there rather than through a separate alert.

Storage / Infrastructure Decision (Frozen)

Same Postgres + pgvector instance as today. No new infrastructure. In-process background jobs (`jobs.py`) are retained — no queue introduced without a demonstrated bottleneck. The depth-3 / breadth-15→10→10 drill cap exists specifically to keep this feasible without one.

Architecture Acceptance Tests (Frozen)

- A niche matching an exclusion rule never appears in any dossier or review queue.
- A niche's `next_recheck_at` is always set on scan completion and honored before any re-scan.
- No two Niche/NicheCandidate rows are created for the same (parent, depth, canonical name) within an active registry window.
- Every Evidence row has both `evidence_type` and `origin` set; no row is created without them.
- A Dossier's evidence trail contains only Evidence rows traceable to a real ResearchRun.
- `research_more` always produces a new ResearchRun at `depth + 1`, never re-uses the same depth.

- Watched dossiers are automatically re-scored within 24 hours of their niche's `next_recheck_at` passing, with no manual action.
- CORP2 tables never hold a foreign key into CORP1's Evidence, NicheCandidate, or ResearchQuery tables — only into Dossier.

Stage 5: Task Decomposition

Architecture is frozen (Stages 3–4) unless a task below exposes a genuine contradiction. This section breaks the complete system into dependency-ordered vertical slices. Coding agents implement this decomposition — they do not redesign the architecture while coding.

The Phase 1–3 sections earlier in this document describe **what** each piece does. This section is the authoritative **build order** and supersedes their numbering for sequencing. It also adds tasks Stage 4 introduced that the original phase list did not cover: the data migration, the Dossier model, the four-state decision gate, the CORP2 handoff package, and the registry re-scan scheduler.

Every task below follows the same acceptance flow: Stage 6 (AI Build) → Stage 7 (Automated Verification — real tests, not claims) → Stage 8 (Independent Review against this architecture) → Stage 9 (Human Gate: **ACCEPT / REVISE / REJECT / ARCHITECTURE REVIEW REQUIRED**) → Stage 10 (Commit & Memory).

Build Order

Task	Title	Depends on	Migration?
T0	Frozen data model migration	none	Yes
T1	Capability interfaces (Phase 1.1)	none	No
T2	Wrap existing adapters (Phase 1.2)	T1	No
T3	Recursive niche discovery engine	T0, T1, T2	No
T4	Campaign pipeline rewire (Phase 1.4)	T3	No
T5	Product idea generation (Phase 1.5)	T3, T4	Yes (ProductIdea table)
T6	Dossier model and generator (Phase 1.6)	T0, T5	No (uses T0's Dossier table)
T7	Scoring engine update (Phase 1.7)	T3	No
T8	Four-state decision gate	T0, T6	No

Task	Title	Depends on	Migration?
T9	CORP2 handoff package	T8	No
T10	Registry re-scan scheduler	T0, T3	No
T11-T15	Phase 2 adapters (2.1-2.5)	T2	No
T16-T21	Phase 3 adapters and full scoring (3.1-3.6)	T2, T7	No

T1 and T0 can start in parallel. T7, T5→T6, and T10 can run in parallel once T3 is accepted. Phase 2/3 adapter tasks (T11+) can each run independently once T2 is accepted — they touch no shared files.

T0 — Frozen Data Model Migration

- **Allowed files:** one new Alembic migration under `corp/alembic/versions/`; `corp/core/models/niche.py`, `niche_candidate.py`, `evidence.py`, `workflow.py`; new `corp/core/models/dossier.py`
- **Forbidden:** no application/pipeline logic changes in this task — schema and ORM models only
- **Migration:** adds `parent_niche_id / depth` to `niches`; `parent_candidate_id / depth` to `niche_candidates`; `evidence_type / origin` to `evidence`; `research_more` enum value + `dossier_id` to `human_decisions`; new `dossiers` and `dossier_evidence` tables; `excluded` added to Niche's `lifecycle_status` enum, `drilling` to NicheCandidate's `status` enum
- **Acceptance criteria:** `alembic upgrade head` runs clean on a fresh DB and the existing dev DB; `alembic downgrade -1` reverses cleanly; every new enum value is reachable from the ORM
- **Tests:** migration up/downgrade round-trip test; model-level test asserting the new columns and enum values exist with correct nullability
- **Human gate:** ACCEPT/REVISE/REJECT — review is migration-safety only, no functional behavior yet

T1 — Capability Interfaces (Phase 1.1)

- **Allowed files:** new `corp/workers/providers/capabilities.py`
- **Forbidden:** no changes to existing adapter files in this task
- **Migration:** none

- **Acceptance criteria:** the `EvidenceProvider` base class and all eight capability interfaces exist with typed method signatures returning structured evidence matching `Evidence.evidence_type` values from T0
- **Tests:** interface contract tests (signature/typing only — no real adapters implement them yet)
- **Human gate:** ACCEPT/REVISE/REJECT

T2 — Wrap Existing Adapters (Phase 1.2)

- **Allowed files:** modify all `corp/workers/adapters/*.py` per the Adapter → Capability mapping already defined in Phase 1.2 above
- **Forbidden:** no changes to existing `collect()` logic — interface methods delegate to it, they do not replace it
- **Migration:** none
- **Acceptance criteria:** every adapter implements the capability interface(s) listed in its Phase 1.2 mapping row; existing adapter tests still pass unmodified
- **Tests:** one conformance test per adapter asserting it satisfies its assigned interface(s)
- **Human gate:** ACCEPT/REVISE/REJECT

T3 — Recursive Niche Discovery Engine

Supersedes Phase 1.3/1.4's single-level drill description with the frozen Stage 3/4 behavior: depth-capped recursion, not a fixed one-level drill.

- **Allowed files:** new `corp/workers/intelligence/niche_discovery.py`; new `rules/niche_discovery_prompt.yaml`
- **Forbidden:** no changes to `corp/api/jobs.py` in this task (that is T4)
- **Migration:** none (T0 already added the columns this needs)
- **Acceptance criteria:**
 - Queries all capability providers in parallel for a broad topic (depth 0)
 - Recurses using the LLM's "can I name a concrete product" stopping rule, hard-capped at depth 3
 - Breadth-capped at 15 candidates at depth 0, 10 per branch at depth 1–2
 - Checks each candidate against `Niche.next_recheck_at` before researching — skips anything inside its 90-day window
 - Matches every candidate against the Stage 3 exclusion list deterministically (not via LLM judgment) and sets `lifecycle_status = excluded` on a match, before it can reach creator discovery
 - Every `NicheCandidate` created sets `parent_candidate_id` and `depth` correctly

- **Tests:** a fixture-driven test asserting depth/breadth caps are enforced; a test asserting an excluded-category candidate never proceeds past exclusion matching; a registry test asserting a candidate inside its recheck window is skipped
- **Human gate:** ACCEPT/REVISE/REJECT/ARCHITECTURE REVIEW REQUIRED — this is the highest-risk task (recursion + cost control), flag for architecture review if caps need adjusting

T4 — Campaign Pipeline Rewire (Phase 1.4)

- **Allowed files:** `corp/api/jobs.py`; `web/src/pages/CampaignDetailPage.tsx`
- **Forbidden:** no changes to `corp/workers/intelligence/niche_discovery.py` (T3) beyond calling it
- **Migration:** none
- **Acceptance criteria:** dispatcher calls T3's discovery engine in place of the old discover/drill-down stages; existing verify/qualify/onboard/research/cluster/score/dossier/review stages remain reachable in sequence
- **Tests:** end-to-end campaign run against a fixture topic reaches a dossier without manual intervention
- **Human gate:** ACCEPT/REVISE/REJECT

T5 — Product Idea Generation (Phase 1.5)

- **Allowed files:** new `corp/core/models/product_idea.py`; new `corp/workers/intelligence/product_ideation.py`; new `rules/product_ideation_prompt.yaml`; one new Alembic migration for the `product_ideas` table
- **Migration:** yes — new `product_ideas` table, FK to problem cluster and creator
- **Acceptance criteria:** 3–5 ideas generated per problem cluster, each with type, complexity, price range, fit rationale; every idea's generation traces to an `Evidence.origin = inference` row for provenance
- **Tests:** generation test against a fixture cluster; provenance test confirming each idea links back to inference evidence
- **Human gate:** ACCEPT/REVISE/REJECT

T6 — Dossier Model and Generator (Phase 1.6)

- **Allowed files:** `corp/workers/dossier/generator.py`; `web/src/pages/CreatorDetailPage.tsx`
- **Forbidden:** no changes to `corp/core/models/dossier.py` (already created in T0)

- **Migration:** none
- **Acceptance criteria:** generator writes to the T0 `Dossier` table (not an ad hoc structure); populates `content`, `dossier_evidence`, and `niche_path` (walking `Niche.parent_niche_id`); includes problem clusters, product ideas, demand validation, opportunity score breakdown, competitive summary, and a confidence-rated recommendation as shown in the Example Dossier Summary above
- **Tests:** a generated dossier's evidence trail contains only rows traceable to a real `ResearchRun` (Stage 4 acceptance test)
- **Human gate:** ACCEPT/REVISE/REJECT

T7 — Scoring Engine Update (Phase 1.7)

- **Allowed files:** `corp/core/scoring/engine.py`; `rules/scoring.yaml`
- **Migration:** none
- **Acceptance criteria:** scoring consumes evidence by `Evidence.evidence_type`, not by source platform; new dimensions (external demand, competitive saturation, purchase intent, audience dissatisfaction) map to the capability types listed in Phase 1.7 above
- **Tests:** scoring unit tests per dimension with fixture evidence sets
- **Human gate:** ACCEPT/REVISE/REJECT

T8 — Four-State Decision Gate

- **Allowed files:** `corp/api/routes_*.py` (decision endpoint), `web/src/pages/CreatorDetailPage.tsx` (or dedicated review UI)
- **Forbidden:** no changes to `HumanDecision` / `Dossier` models (T0 owns schema)
- **Migration:** none
- **Acceptance criteria:** UI/API exposes exactly Reject / Research More / Watch / Approve; **Research More** creates a new `ResearchRun` at `depth + 1` via T3's engine; **Watch** sets `Dossier.status = watching` with no other side effect; **Approve** sets `Dossier.status = approved` and triggers T9
- **Tests:** one test per decision outcome asserting the correct downstream effect and that `HumanDecision` remains append-only
- **Human gate:** ACCEPT/REVISE/REJECT

T9 — CORP2 Handoff Package

- **Allowed files:** new `corp/workers/handoff/corp2_export.py`
- **Forbidden:** no direct writes to any CORP2 (Acq*-prefixed) table — CORP1 never reaches across the boundary; it produces a package CORP2 pulls or receives via API

- **Migration:** none
- **Acceptance criteria:** package contains the full dossier, complete evidence trail, full niche drill-down path, and the reviewer's decision-gate notes, matching the Stage 3 Handoff Package definition exactly
- **Tests:** a test asserting the package is fully reconstructable from `Dossier` + `dossier_evidence` + `Niche.parent_niche_id` walk alone, with no other CORP1 table access required
- **Human gate:** ACCEPT/REVISE/REJECT

T10 — Registry Re-scan Scheduler

- **Allowed files:** new `corp/workers/scheduler/registry_rescan.py`
- **Migration:** none
- **Acceptance criteria:** any `Dossier.status = watching` whose `Niche.next_recheck_at` has passed is automatically re-scored within 24 hours, satisfying the Stage 4 acceptance test; runs as an in-process scheduled job, no new queue infrastructure
- **Tests:** a time-travel test (mocked clock) asserting a watched dossier past its recheck date gets re-scored and a fresh one does not
- **Human gate:** ACCEPT/REVISE/REJECT

T11-T21 — Phase 2 and Phase 3 Adapters

Each adapter task (Pinterest, Google Search/Autocomplete, Kickstarter/Indiegogo, Patreon/Substack, Quora, Google Books, Google Shopping, Trustpilot, Product Hunt, plus T21 Full Scoring Integration) follows the same template:

- **Allowed files:** one new adapter file under `corp/workers/adapters/`, implementing the capability interface(s) listed in its Phase 2/3 entry above
- **Forbidden:** no changes to any other adapter or to the scoring engine (T21 is the sole exception — it updates `rules/scoring.yaml` weighting once all sources exist)
- **Migration:** none
- **Acceptance criteria:** conformance test against its assigned capability interface(s); a live-fixture test confirming it collects the evidence fields listed in its Phase 2/3 description
- **Human gate:** ACCEPT/REVISE/REJECT per adapter — these can be reviewed and merged independently and in any order

The CORP1 Pipeline

The pipeline runs in 9 steps. You provide a broad topic. CORP1 delivers ranked opportunities with dossiers.

Step 1: Discover Broad Topics and Drill Into Niches

The system discovers niches autonomously from data. You can also manually suggest a niche to research — but you never have to.

Level 0 — Find broad topics. The system scans Google Trends (and other trend sources) to identify broad topic areas with current momentum — e.g., "small business", "home organization", "personal finance".

Level 1 — Identify main niches. For each broad topic, the system queries all sources in parallel. Each source reveals a different angle:

- **Google Trends** → demand signals and direction
- **Reddit** → pain points people discuss
- **YouTube search** → what content creators already make
- **Marketplaces (Gumroad, Etsy, Udemy)** → what people already pay for

The LLM reads the raw data and organises what it finds into main niches (e.g., under "Automotive": DIY Maintenance, Detailing, Track Builds, Restoration, etc.).

Level 2+ — Recursive drill-down. Some niches are still too broad to build a digital product around. "Track Builds" — what product would you make? The system goes back to the data sources with a more specific query and drills deeper: Aero, Suspension, Tyres. Now you can picture the product — a suspension setup calculator, a tyre compound guide.

Stopping rule: At each level the LLM asks: "Is this specific enough to name a concrete digital product someone would pay for?" If yes, it's a valid leaf niche. If no, drill deeper with another round of data queries.

The niches come from the evidence, not from the LLM's imagination. Every level is driven by querying real data sources. Your only role is at the decision gate (Step 9) — approve, reject, or request more research.

Research registry. The system maintains a registry of every broad topic, niche, and sub-niche it has already researched — including the date, sources queried, and depth reached. Before starting any new discovery pass, it checks the registry to avoid repeating work. If Google Trends surfaces "small business" again next month, the system knows it already drilled that tree and skips it (unless enough time has passed that a re-scan is warranted, or new data sources have been added).

- Small business bookkeeping tools

- Local restaurant marketing
- Freelance graphic design tools
- Home cleaning service management
- E-commerce shipping for small sellers

Step 2: Drill Down

For each niche the LLM drills one level deeper — niche within a niche. "Small business bookkeeping tools" becomes "Cloud bookkeeping for solo contractors". This is where the real opportunities hide.

Step 3: Find Creators

For each drilled-down niche, search for creators across YouTube, Instagram, TikTok, Pinterest, and other platforms. Filter to the sweet spot: 10,000–200,000 followers. Large enough to have a buying audience. Small enough for a realistic partnership.

Step 4: Build Creator Profile

For each creator, automatically collect:

- What niche they operate in
- What they teach or talk about
- What products or services they already promote
- Posting frequency and content themes
- Audience size and engagement signals
- Website, newsletter, store, or other monetisation channels

Step 5: Research Content and Audience

This is where CORP1 does its deepest work.

Content research — examine video titles, descriptions, transcripts, posts, captions, blog content, Pinterest boards, and search results around the creator.

Audience research — examine comments, questions, Reddit discussions, Google search demand, Pinterest search activity, YouTube comments, FAQs, and community discussions. Extract signals such as:

- "How do I..."
- "Where can I find..."
- "Can you make a..."
- "I wish there was..."

- "What template do you use?"

Step 6: Convert Signals to Problem Clusters

Group hundreds of audience comments and questions into problem clusters. Instead of giving you 600 comments, CORP1 reports:

Problem Cluster: Garden Planning — 143 related audience signals, high recurrence, strong purchase intent, existing commercial products found.

Step 7: Generate Product Ideas

For each strong problem cluster, propose specific products that fit the creator's brand and audience:

- Templates and calculators
- Notion systems and databases
- Guides and checklists
- Courses and memberships
- Apps and trackers
- AI-powered tools

Step 8: Validate Demand and Rank

Search outside the creator's audience for external evidence:

- Google search demand and trends
- Pinterest trends and activity
- Reddit discussions
- Existing products on Etsy, Gumroad, Amazon, Udemy
- Competitor analysis
- Advertiser spending around the problem
- Kickstarter and Indiegogo backing signals
- Patreon and Substack monetisation signals

Then score each opportunity by: problem frequency, audience engagement, purchase intent, market demand, competition, product complexity, expected price, creator fit, build ease, and recurring revenue potential.

Step 9: Human Decision Gate

CORP1 presents a dossier for each strong opportunity. You decide:

Reject → Research More → Approve

CORP1 never acts without your approval. It brings the strongest opportunities to you. You decide what happens next.

Evidence Sources

Each data source proves something different. CORP1 gathers evidence across all of them to build a complete picture.

Problem Evidence — People have this problem

Source	What it provides
YouTube comments	Direct audience questions and frustrations
Reddit discussions	Unfiltered community pain points
Stack Exchange	Technical problems with no good solution
Quora	Broad audience questions seeking answers

Search Intent Evidence — People actively look for answers

Source	What it provides
Google Search and Autocomplete	What people type into Google
Google Trends	Demand direction over time
Pinterest Trends	Visual and planning-oriented demand

Planning Intent Evidence — People are researching and saving solutions

Source	What it provides
Pinterest boards and pins	People saving and organising solutions
YouTube playlists and saves	Content people return to

Transaction Evidence — Solutions exist and people pay for them

Source	What it provides
Etsy	Handmade and digital product sales
Amazon	Physical and digital product market
Gumroad	Creator-sold digital products
Udemy	Course and educational content market
App Store	Mobile app solutions
Product Hunt	New product launches
Google Books	Published knowledge products
Google Shopping	Product and pricing intelligence

Commercial Intent Evidence — People will pay real money

Source	What it provides
Kickstarter and Indiegogo	People pre-ordering and backing solutions
Patreon	People paying creators for ongoing content
Substack	People paying for written expertise

Dissatisfaction Evidence — Current solutions are not good enough

Source	What it provides
Amazon reviews	What people dislike about existing products
App Store reviews	What is broken in current apps
Trustpilot	Service and product complaints
Reddit complaints	Unfiltered frustration with solutions

The Google Breakdown

A single Google query fans out into five different evidence types:

Google Product	Evidence Type
Google Trends	TREND — demand direction
Search intelligence and Autocomplete	SEARCH_INTENT — what people look for
YouTube	PROBLEM and RECURRENCE — audience pain
Google Books	SOLUTION — published answers
Google Shopping	SOLUTION and COMPETITION — what exists and what it costs

Capability Interface Architecture

The scoring engine should not care which platform evidence came from. It should care what KIND of evidence it is.

The Problem with Platform-Specific Logic

If the scoring engine contains logic like "YouTube comments mean problems" and "Etsy listings mean competition", then every new platform requires changes to the scoring code. This does not scale.

The Solution: Evidence Providers

Instead, each adapter implements one or more capability interfaces:

EvidenceProvider (base)

- **ProblemProvider** — surfaces audience pain points
- **SearchIntentProvider** — surfaces what people search for
- **TrendProvider** — surfaces demand direction over time
- **PlanningIntentProvider** — surfaces research and saving behaviour
- **TransactionProvider** — surfaces existing purchases and pricing
- **SolutionProvider** — surfaces existing products and competitors
- **MonetisationProvider** — surfaces creator payment signals
- **DissatisfactionProvider** — surfaces complaints about current solutions

How Platforms Map to Capabilities

One platform can implement multiple interfaces:

Platform	Capabilities
YouTube	ProblemProvider, TrendProvider
Reddit	ProblemProvider, DissatisfactionProvider
Stack Exchange	ProblemProvider
Quora	ProblemProvider
Google Trends	TrendProvider
Google Search	SearchIntentProvider
Pinterest	PlanningIntentProvider, TrendProvider
Etsy	TransactionProvider, SolutionProvider
Amazon	TransactionProvider, SolutionProvider, DissatisfactionProvider
Gumroad	TransactionProvider, SolutionProvider
Udemy	TransactionProvider, SolutionProvider
App Store	SolutionProvider, DissatisfactionProvider
Product Hunt	SolutionProvider
Google Books	SolutionProvider
Google Shopping	SolutionProvider, TransactionProvider
Kickstarter	TransactionProvider
Indiegogo	TransactionProvider
Patreon	MonetisationProvider
Substack	MonetisationProvider
Trustpilot	DissatisfactionProvider

How the Scoring Engine Consumes Evidence

The scoring engine asks for evidence by type, not by platform:

- "Give me all ProblemEvidence" → YouTube, Reddit, Stack Exchange, and Quora all respond
- "Give me all TrendEvidence" → Google Trends and Pinterest respond
- "Give me all TransactionEvidence" → Etsy, Gumroad, Kickstarter respond

- "Give me all DissatisfactionEvidence" → Amazon reviews, App Store, Trustpilot respond

Adding Pinterest later does not change the scoring code. It just registers as a `PlanningIntentProvider` and a `TrendProvider`.

Benefits

1. **Clean scoring** — scores by evidence type, not platform
2. **Easy to extend** — new platform = implement the right interfaces
3. **Composable** — one platform provides multiple evidence types
4. **Testable** — mock any provider without mocking a platform

What Is Already Built

CORP1 has a working codebase with significant infrastructure in place. The challenge is not starting from scratch — it is rewiring the existing pieces into the correct pipeline.

Working Adapters (12 platforms)

Adapter	Lines	Status
YouTube (API + yt-dlp)	475 + 120	Working — videos, comments, captions, search
Reddit	130	Working — posts, comment trees
Stack Exchange	180	Working — questions, answers, pagination
Google Trends	300	Working — interest over time, related queries, news
Marketplace (Gumroad, Etsy, Udemy)	520	Working — product listings, prices, reviews
Amazon Reviews	150	Working — review extraction and analysis
App Store	140	Working — app listings and reviews
HackerNews	130	Working — stories and discussions
Wikipedia	140	Working — article summaries, page views

Adapter	Lines	Status
Web Presence	220	Working — website detection and metadata
Search Demand	180	Working — keyword expansion and intent
Google Trends (RSS)	included above	Working — trending stories

Working Intelligence Pipeline

Component	Lines	What it does
Extraction pipeline	417	LLM-powered observation extraction from content
Clustering engine	179	HDBSCAN-based problem cluster detection
Cluster pipeline	230	End-to-end cluster orchestration
Scoring pipeline	537	10-dimension opportunity scoring from YAML rules
Scoring engine	189	Score computation, hashing, and band assignment
Topic classification	160	LLM-powered topic extraction
Niche selection and qualification	280	Rule-based niche filtering and ranking
Dossier generator	271	Structured dossier output for human review

Working Infrastructure

- **FastAPI backend** with async SQLAlchemy, PostgreSQL 16 with pgvector
- **React 19 dashboard** with TanStack Query, campaign pipeline UI, creator detail pages, decision panel
- **LLM providers:** Gemini, Groq, FAIR (free AI router with quality gating)
- **Background job system** with status tracking and polling
- **Alembic migrations** for database schema management
- **Human decision gate** — approve, reject, or watch with rationale
- **836 passing tests** with full mypy strict compliance

Adapters Not Yet Built

Adapter	Evidence Type	Priority
Pinterest	Planning intent, trends	High
Google Search and Autocomplete	Search intent	High
Quora	Problem evidence	Medium
Google Books	Solution evidence	Medium
Google Shopping	Competition and pricing	Medium
Kickstarter and Indiegogo	Commercial intent	Medium
Patreon	Monetisation signals	Medium
Substack	Monetisation signals	Medium
Trustpilot	Dissatisfaction evidence	Low
Product Hunt	Solution evidence	Low

Phase 1: Rewire Pipeline and Capability Interfaces

Goal: get the end-to-end pipeline working with existing adapters. User enters a broad topic and receives ranked opportunities with dossiers.

1.1 Define Capability Interfaces

Create the EvidenceProvider base class and the seven capability interfaces (ProblemProvider, SearchIntentProvider, TrendProvider, PlanningIntentProvider, TransactionProvider, SolutionProvider, MonetisationProvider, DissatisfactionProvider). Each interface defines a standard method signature that returns typed evidence.

Files: new corp/workers/providers/capabilities.py

1.2 Wrap Existing Adapters

Make each of the 12 existing adapters implement the appropriate capability interfaces. No rewrite — add the interface methods that delegate to existing collect logic.

Adapter	Implements
YouTube	ProblemProvider

Adapter	Implements
Reddit	ProblemProvider, DissatisfactionProvider
Stack Exchange	ProblemProvider
Google Trends	TrendProvider
Marketplace	TransactionProvider, SolutionProvider
Amazon Reviews	DissatisfactionProvider
App Store	SolutionProvider, DissatisfactionProvider
HackerNews	ProblemProvider
Search Demand	SearchIntentProvider
Web Presence	(unchanged — profile enrichment)

Files: modify all corp/workers/adapters/*.py

1.3 Build LLM Niche Discovery Stage

Replace the current scrape-first discovery with an LLM-driven approach:

1. Query Google Trends, Reddit, YouTube search, and Marketplace in parallel for the broad topic
2. Feed all results to the LLM with a prompt: "Given this evidence about the topic, identify 10–15 specific niches"
3. LLM returns structured niche list with reasoning
4. For each niche, LLM drills down one level into a sub-niche
5. Store as NicheCandidate rows linked to the campaign

Files: new corp/workers/intelligence/niche_discovery.py, new rules/niche_discovery_prompt.yaml

1.4 Rewire Campaign Pipeline Stages

Update the campaign job dispatcher to use the new flow:

1. **discover** → LLM niche discovery (new)
2. **drill-down** → LLM sub-niche generation (new)
3. **validate** → check each sub-niche against real data (reuse existing verify and qualify)
4. **find-creators** → search for creators in validated sub-niches (reuse existing onboarding)
5. **research** → run full content and audience research per creator (existing pipeline)

- 6. **cluster** → problem clustering (existing)
- 7. **score** → opportunity scoring (existing)
- 8. **dossier** → generate dossier (existing)
- 9. **review** → human decision gate (existing)

Files: modify corp/api/jobs.py, update web/src/pages/CampaignDetailPage.tsx

1.5 Build Product Idea Generation

New LLM-powered stage that takes problem clusters and generates product ideas:

- Input: problem cluster with evidence, creator profile, audience signals
- Output: 3–5 product ideas per cluster, each with type (template, course, app, etc.), estimated complexity, price range, and fit rationale
- Stored as ProductIdea rows linked to the creator and cluster

Files: new corp/core/models/product_idea.py, new corp/workers/intelligence/product_ideation.py, new rules/product_ideation_prompt.yaml

1.6 Enhance the Dossier

Update the dossier generator to include:

- Problem clusters with evidence counts and recurrence
- Product ideas with complexity and price estimates
- External demand validation summary
- Opportunity score with component breakdown
- Competitive landscape summary
- Clear recommendation with confidence level

Files: modify corp/workers/dossier/generator.py, update web/src/pages/CreatorDetailPage.tsx

1.7 Update Scoring Engine

Modify the scoring engine to consume evidence by capability type rather than by platform. Add new scoring dimensions:

- External demand strength (from TrendProvider and SearchIntentProvider evidence)
- Competitive saturation (from SolutionProvider evidence)
- Purchase intent (from TransactionProvider evidence)
- Audience dissatisfaction (from DissatisfactionProvider evidence)

Files: modify corp/core/scoring/engine.py, update rules/scoring.yaml

Phase 1 Deliverable

A user enters "small business", and CORP1:

1. Discovers niches using existing data sources and LLM synthesis
2. Drills down to sub-niches
3. Finds creators in those sub-niches
4. Researches their content and audience
5. Clusters problems and generates product ideas
6. Scores opportunities
7. Presents a dossier with a clear recommendation
8. Waits for human decision

Phase 2: High-Value New Adapters

Goal: fill the biggest evidence gaps by adding the most impactful data sources.

2.1 Pinterest Adapter

Why it matters: Pinterest is where people plan and save solutions. A niche with high Pinterest activity signals strong planning intent — these are people actively preparing to buy or build.

Capabilities: PlanningIntentProvider, TrendProvider

What it collects:

- Pin search results for niche keywords
- Board themes and organisation patterns
- Trending pins and topics
- Save counts as demand signals
- Related search suggestions

2.2 Google Search and Autocomplete Adapter

Why it matters: Google Autocomplete reveals what millions of people actually type. "Small business bookkeeping" autocompleting to "small business bookkeeping template free" is direct evidence of demand for a specific product type.

Capabilities: SearchIntentProvider

What it collects:

- Autocomplete suggestions for niche keywords

- "People also ask" questions
- Related searches
- Search result analysis (what ranks, what is missing)
- Featured snippet content

2.3 Kickstarter and Indiegogo Adapter

Why it matters: Someone backing a Kickstarter project has demonstrated the strongest form of commercial intent — they paid money for something that does not exist yet. This is the most powerful demand signal available.

Capabilities: TransactionProvider

What it collects:

- Projects in the niche category
- Funding amounts and backer counts
- Project descriptions and categories
- Comments and community engagement
- Success and failure rates

2.4 Patreon and Substack Adapter

Why it matters: If people already pay creators in a niche for content, that niche has proven monetisation potential. A creator with Patreon supporters has an audience willing to spend money.

Capabilities: MonetisationProvider

What it collects:

- Creator pages in the niche
- Tier structures and pricing
- Patron and subscriber counts where visible
- Content themes and offerings
- Community activity signals

2.5 Quora Adapter

Why it matters: Quora questions are structured problem statements. "What is the best way to track inventory for a small bakery?" is a direct problem signal with implicit product demand.

Capabilities: ProblemProvider

What it collects:

- Questions in the niche
- Answer counts and quality
- View counts as demand signals
- Related questions
- Topic followers

Phase 2 Deliverable

CORP1 now gathers evidence from 17 sources instead of 12. The scoring engine receives planning intent, search intent, commercial intent, and monetisation signals that were previously invisible. Opportunity rankings become significantly more accurate.

Phase 3: Complete Evidence Coverage

Goal: fill the remaining gaps for comprehensive opportunity validation.

3.1 Google Books Adapter

Capabilities: SolutionProvider

Search Google Books for published material in each niche. Books on a topic confirm it has enough depth and demand for knowledge products. The absence of books in a niche can signal an underserved opportunity.

3.2 Google Shopping Adapter

Capabilities: SolutionProvider, TransactionProvider

Query Google Shopping for products in each niche. Reveals pricing, number of sellers, product categories, and advertiser spending. This is the strongest competitive landscape signal for physical and digital products.

3.3 Trustpilot Adapter

Capabilities: DissatisfactionProvider

Search Trustpilot for companies in each niche. Extract negative reviews to identify where existing solutions fail. Dissatisfaction with current products is an opportunity signal.

3.4 Product Hunt Adapter

Capabilities: SolutionProvider

Search Product Hunt for launches in each niche. Reveals what entrepreneurs are building, community reception, and feature gaps. Recent launches confirm market activity; poor reception confirms unmet needs.

3.5 Indiegogo Adapter

Capabilities: TransactionProvider

Same approach as Kickstarter. Indiegogo tends to have more international and hardware-adjacent projects, broadening the commercial intent signal.

3.6 Full Scoring Integration

With all evidence providers in place, update the scoring engine to weight evidence across all dimensions:

Scoring Dimension	Evidence Sources
Problem frequency	YouTube, Reddit, SE, Quora, HackerNews
Search demand	Google Search, Autocomplete
Trend direction	Google Trends, Pinterest Trends
Planning activity	Pinterest
Purchase intent	Kickstarter, Indiegogo, marketplace transactions
Competition level	Etsy, Gumroad, Amazon, Google Shopping, Product Hunt
Solution gaps	Amazon reviews, App Store reviews, Trustpilot, Reddit
Monetisation proof	Patreon, Substack
Knowledge depth	Google Books, Udemy courses
Creator-audience fit	Content analysis, engagement patterns

Phase 3 Deliverable

CORP1 gathers evidence from 22 or more sources across 8 evidence dimensions. Every opportunity score is backed by multi-source validation. The dossier provides a comprehensive, evidence-based recommendation for each creator opportunity.

The Dossier

The dossier is what CORP1 delivers to you for each strong opportunity. It is the single document that lets you make an informed decision without doing any manual research.

What the Dossier Contains

Creator Profile

- Name, platforms, follower counts
- Niche and content themes
- Posting frequency and engagement rates
- Existing products and monetisation channels
- Website, newsletter, and store links

Audience Analysis

- Top audience questions and pain points (grouped)
- Recurring themes with frequency counts
- Language patterns ("How do I", "I wish there was")
- Engagement quality indicators

Problem Clusters

For each detected problem:

- Problem name and description
- Number of related audience signals
- Recurrence rating (one-off vs recurring)
- Purchase intent strength
- Sample evidence (actual comments and questions)

Product Ideas

For each strong problem cluster:

- Recommended product type (template, course, app, tracker, etc.)
- Why this product fits this creator's audience
- Estimated build complexity (Low, Medium, High)
- Suggested price range
- Comparable products that already exist

Demand Validation

- Google search volume and trend direction

- Pinterest activity and save counts
- Reddit discussion frequency
- Marketplace competitors (Etsy, Gumroad, Amazon, Udemy)
- Kickstarter or Indiegogo backing signals
- Patreon or Substack monetisation signals

Opportunity Score

- Overall score with confidence band (High, Moderate, Emerging)
- Component breakdown: problem frequency, engagement, intent, demand, competition, complexity, price, fit, build ease, recurring revenue
- Evidence count per dimension

Competitive Landscape

- Direct competitors (similar products already sold)
- Substitute products (different format, same problem)
- Price range of existing solutions
- Review sentiment of existing solutions
- Gaps and weaknesses in current offerings

Recommendation

- Clear recommendation with confidence level
- Strongest evidence summary
- Key risks and unknowns
- Suggested next steps

Example Dossier Summary

Creator A — Opportunity: Meal Planning System

- 287 relevant audience questions detected
- Strong Google and Pinterest demand
- Moderate competition (3 existing products, average rating 3.2 stars)
- Estimated build complexity: Low
- Possible price: 19 to 39 dollars
- Evidence confidence: High
- Recommendation: Approve for outreach

Your Decision

For each dossier you choose one of four options (see Human Decision Gate above for full behavior):

- **Reject** — not a strong enough opportunity
- **Research More** — promising but need more evidence
- **Watch** — promising, not ready; auto re-scored every 90 days
- **Approve** — move to CORP2 for outreach and partnership